HOTS

1. If the iodine step were omitted in the Gram's staining procedure, what colour would you expect Gram positive and Gram negative bacteria to stain?
   a. Gram positive : pink and
      Gram negative : purple
   b. Gram positive : purple and
      Gram negative : pink
   c. Gram positive : purple and
      Gram negative : purple
   d. Gram positive : pink and
      Gram negative : pink
2. In a Gram's staining method, a step could be omitted and still allow differentiation between Gram positive and Gram negative cells. Name the step.

3.7.4 Importance of Gram Staining

This century old staining method still remains as the universal basis for bacterial classification and identification. Even with today's elaborate and expensive medical technology, the Gram's staining remains an important, inexpensive and unbeatable tool in the identification of pathogens.

Examination of Gram stained organisms usually provides the basis for classifying, identifying and characterizing bacteria. Gram staining of clinical specimens, however provides only a preliminary indication of the identity of the etiological agent (the organism causing the disease). Gram nature of common pathogenic bacteria is given in Table 3.4.

Gram stains of clinical specimens or of growth on culture plates are especially important in determining the most effective antibiotic for the ill patients who required immediate therapy.

Infobits

There are several modifications of Gram's Stain

- Kopeloff and Beerman's modification.
- Jensen's modification.
- Weigert's modification.
- Preston and Morell's modification.

Prof. Hans Christian Gram
(September 13, 1853-November 14, 1938)



In 1884, Prof. Hans Christian Gram while examining lung tissue from patients who had died of pneumonia, discovered that certain stains were preferentially taken up and retained by bacterial cells. Gram was a modest man, and in his initial publication he remarked, "I have therefore published the method, although I am aware that as yet it is very defective and imperfect; but it is hoped that also in the hands of other investigators it will turn out to be useful". Dr. Gram used Bismarck brown instead of Safranin. It was a few years later, German pathologist Carl Weigert (1845-1904), added the final step of staining with Safranin.

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